

CS3010 – Computer Vision

Transformations

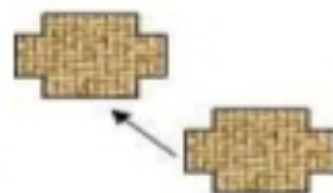
- ▶ Process – converts object in to another form
- ▶ Change in form, nature or appearance
- ▶ Types of transformation
 - ▶ Translation
 - ▶ Reflection
 - ▶ Rotation
 - ▶ Dilation

Types of Transformations

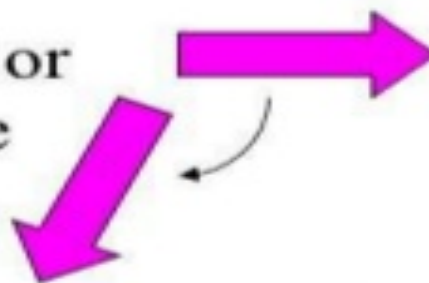
Reflections: These are like mirror images as seen across a line or a point.



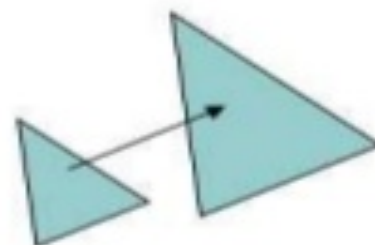
Translations (or slides): This moves the figure to a new location with no change to the looks of the figure.



Rotations: This turns the figure clockwise or counter-clockwise but doesn't change the figure.

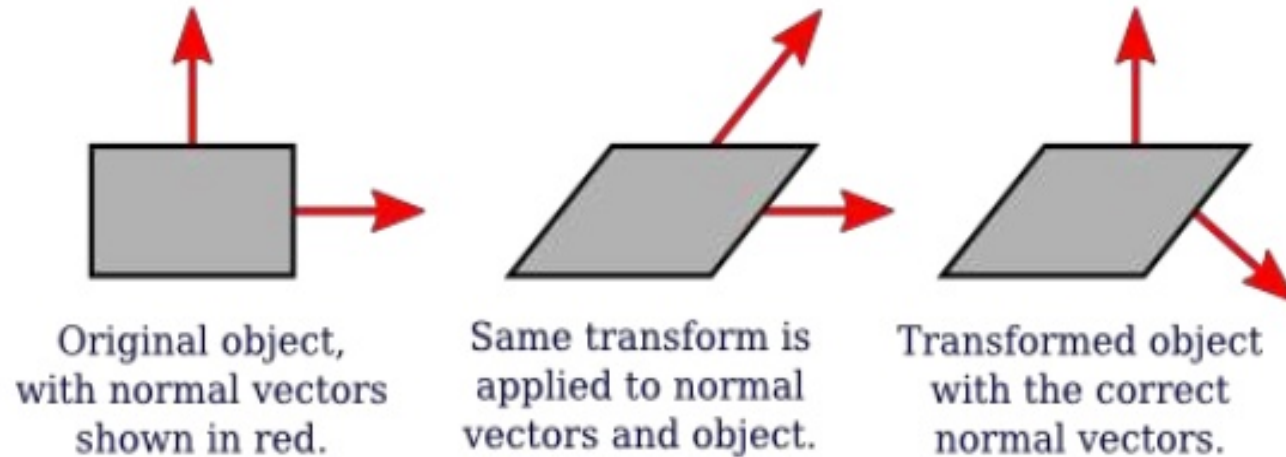


Dilations: This reduces or enlarges the figure to a similar figure.



Geometric Transformations

- ▶ Changes the orientation , size and shape of the objects
- ▶ Alter the coordinate descriptions of objects

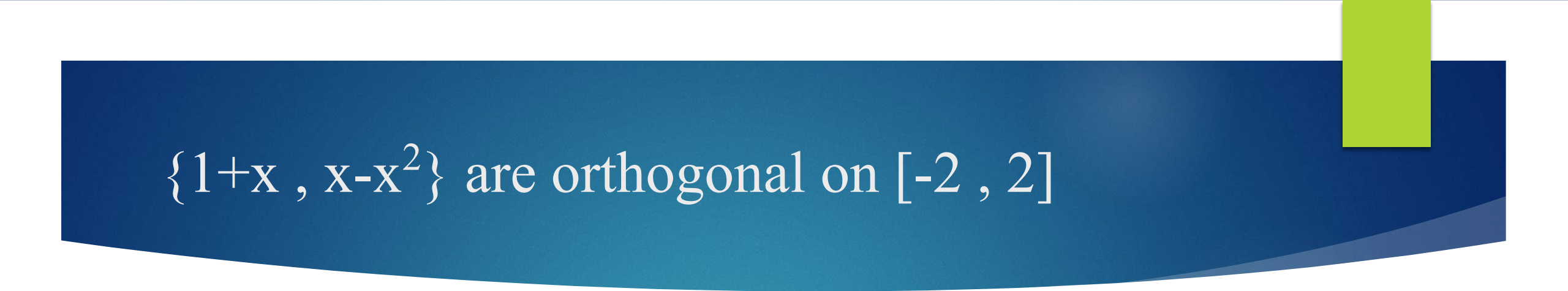


Linear transformation

- ▶ Converts one type of data to another type of data
- ▶ Its of the form $y = A x$
- ▶ A – is called standard matrix
- ▶ $|A| = 0$ then A is called Singular
- ▶ $|A| \neq 0$ then A is called Non Singular or Regular
- ▶ Inverse Transformation $x = A^{-1} y$

Inner Product

- Consider two functions $f(x)$ and $g(x)$ on the interval $[a, b]$
- ▶ Then Inner product of f and g is given as (f, g)
- ▶ $(f, g) = \int_a^b f(x)g(x)dx$
- ▶ If $(f, g) = 0$, we say f and g are orthogonal



$\{1+x, x-x^2\}$ are orthogonal on $[-2, 2]$

- ▶ Inner product is zero
- ▶ Two functions are orthogonal in nature

Orthogonal Transformation

- ▶ It is a linear transformation $T: V \rightarrow V$ which preserves a symmetric inner product
- ▶ It preserves the lengths of vectors and angles between vectors

Euclidean Transformation

- ▶ The Euclidean transformations are the most commonly used transformations.
- ▶ An Euclidean transformation is either a translation, a rotation, or a reflection.

Translations and Rotations on the xy -Plane

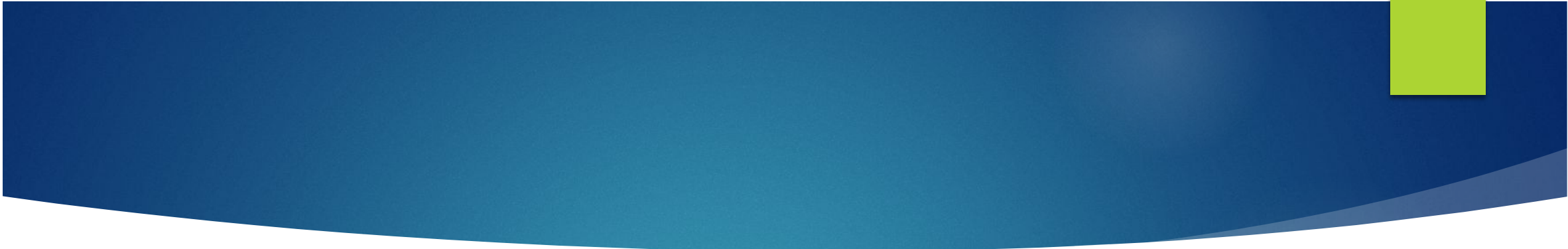
- ▶ We intend to translate a point in the xy -plane to a new place by adding a vector $\langle h, k \rangle$.
- ▶ It is not difficult to see that between a point (x, y) and its new place (x', y') ,
- ▶ we have $x' = x + h$ and $y' = y + k$.
- ▶ Let us use a form similar to the homogeneous coordinates.
- ▶ That is, a point becomes a column vector whose third component is 1. Thus, point (x, y) becomes

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

- Then, the relationship between (x, y) and (x', y') can be put into a matrix form like

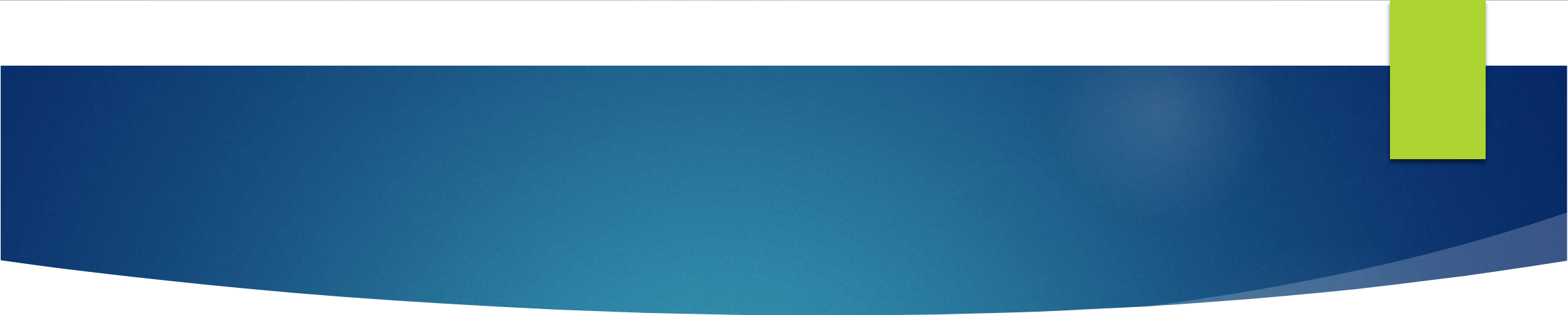
$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & h \\ 0 & 1 & k \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -h \\ 0 & 1 & -k \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$

- 
- ▶ Therefore, if a line has an equation $Ax + By + C = 0$,
 - ▶ after plugging the formulae for x and y ,
 - ▶ the line has a new equation
 - ▶ $Ax' + By' + (-Ah - Bk + C) = 0$

- If a point (x, y) is rotated an angle a about the coordinate origin to become a new point (x', y') , the relationships can be described as

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos a & -\sin a & 0 \\ \sin a & \cos a & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} \cos a & \sin a & 0 \\ -\sin a & \cos a & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$



Thus, rotating a line $Ax + By + C = 0$ about the origin a degree brings it to a new equation:

$$(A\cos a - B\sin a)x' + (A\sin a + B\cos a)y' + C = 0$$

Combination of translation and rotation

- ▶ Translations and rotations can be combined into a single equation
- ▶ Rotate the point (x,y) about an angle a about the coordinate origin and translates the rotated result in the direction of (h,k)

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos a & -\sin a & h \\ \sin a & \cos a & k \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} \cos a & \sin a & -h \cos a - k \sin a \\ -\sin a & \cos a & h \sin a - k \cos a \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$

- ▶ if translation (h,k) is applied first followed by a rotation of angle a (about the coordinate origin)

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos a & -\sin a & h \cos a - k \sin a \\ \sin a & \cos a & h \sin a + k \cos a \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

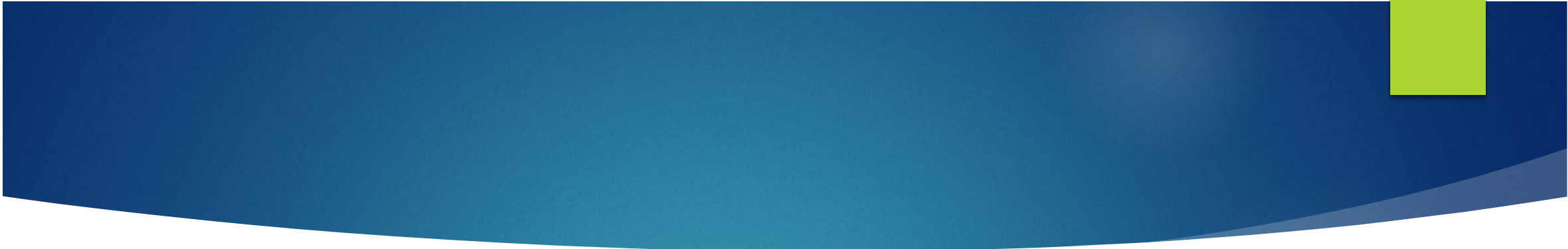
$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} \cos a & \sin a & -h \\ -\sin a & \cos a & -k \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$

Therefore, rotation and translation are not commutative!

Translations and Rotations in Space

- translates points by adding a vector $\langle p, q, r \rangle$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & p \\ 0 & 1 & 0 & q \\ 0 & 0 & 1 & r \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \qquad \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & -p \\ 0 & 1 & 0 & -q \\ 0 & 0 & 1 & -r \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix}$$

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- ▶ Rotations in space are more complex
 - ▶ Because we can either rotate about the x -axis, the y -axis or the z -axis.
 - ▶ When rotating about the z -axis, only coordinates of x and y will change and the z -coordinate will be the same.
 - ▶ In effect, it is exactly a rotation about the origin in the xy -plane.

Rotation about z axis 90 degrees

x-axis rotates to the y-axis and the y-axis rotates to the negative direction of the original x-axis.

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos a & -\sin a & 0 & 0 \\ \sin a & \cos a & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \quad \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} \cos a & \sin a & 0 & 0 \\ -\sin a & \cos a & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix}$$

Rotation about X axis

y -axis rotates to the z -axis and the z -axis rotates to the negative direction of the original y -axis.

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos a & -\sin a & 0 \\ 0 & \sin a & \cos a & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \quad \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos a & \sin a & 0 \\ 0 & -\sin a & \cos a & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix}$$

Rotation about y axis

x -axis rotates to the negative direction of the z -axis and the z -axis rotates to the original x -axis.

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos a & 0 & \sin a & 0 \\ 0 & 1 & 0 & 0 \\ -\sin a & 0 & \cos a & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \quad \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} \cos a & 0 & -\sin a & 0 \\ 0 & 1 & 0 & 0 \\ \sin a & 0 & \cos a & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix}$$

- ▶ A rotation matrix and a translation matrix can be combined into a single matrix as follows,
- ▶ where the r 's in the upper-left 3-by-3 matrix form a rotation and p , q and r form a translation vector.
- ▶ This matrix represents rotations followed by a translation.

$$\mathbf{x}' = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p \\ r_{21} & r_{22} & r_{23} & q \\ r_{31} & r_{32} & r_{33} & r \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \mathbf{x}$$

Euclidean Transformations

- ▶ Euclidean transformations preserve length and angle measure.
- ▶ Moreover, the shape of a geometric object will not change.
- ▶ That is, lines transform to lines, planes transform to planes, circles transform to circles, and ellipsoids transform to ellipsoids.
- ▶ Only the position and orientation of the object will change.

Affine Transformations

- * Combines linear transformations, and Translations
- * Properties
 - * Origin does not necessarily map to origin
 - * Lines map to lines
 - * Parallel lines remain parallel
 - * Ratios are preserved



$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

Affine Transformations

- ▶ Affine transformations are generalizations of Euclidean transformations.
- ▶ Under affine transformations, lines transform to lines; but, circles become ellipses.
- ▶ Length and angle are not preserved.

Scaling

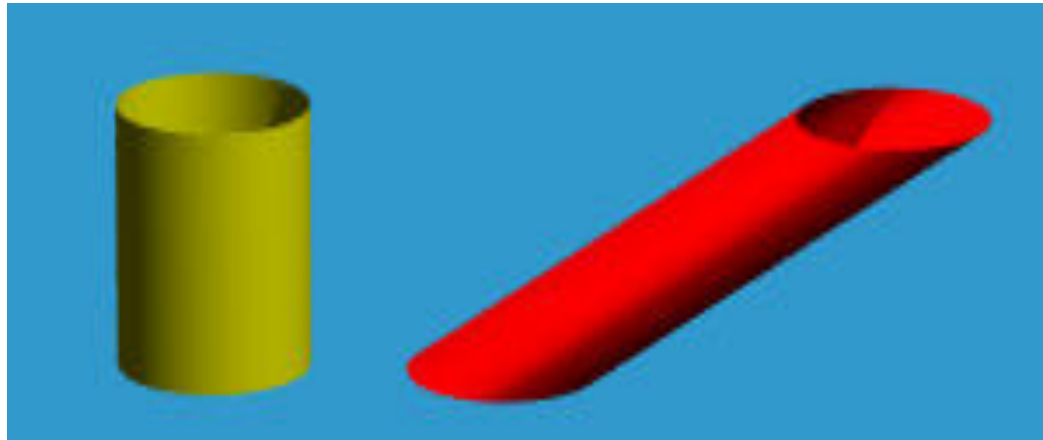
- ▶ Scaling transformations stretch or shrink a given object and, as a result, change lengths and angles.
- ▶ So, scaling is not an Euclidean transformation.
- ▶ The meaning of scaling is making the new scale of a coordinate direction p times larger.
- ▶ In other words, the x coordinate is "enlarged" p times.
- ▶ This requirement satisfies $x' = p x$ and therefore $x = x'/p$.

- Scaling can be applied to all axes, each with a different *scaling factor*.
- For example, if the x -, y - and z -axis are scaled with scaling factors p , q and r , respectively, the transformation matrix is:

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} p & 0 & 0 & 0 \\ 0 & q & 0 & 0 \\ 0 & 0 & r & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \quad \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} 1/p & 0 & 0 & 0 \\ 0 & 1/q & 0 & 0 \\ 0 & 0 & 1/r & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix}$$

Shearing

- ▶ The effect of a shear transformation looks like ``pushing" a geometric object in a direction parallel to a coordinate plane (3D) or a coordinate axis (2D).
- ▶ In the following, the red cylinder is the result of applying a shear transformation to the yellow cylinder:



- ▶ How far a direction is pushed is determined by a *shearing factor*.
- ▶ On the xy -plane, one can push in the x -direction, positive or negative, and keep the y -direction unchanged.
- ▶ Or, one can push in the y -direction and keep the x -direction fixed.
- ▶ The following is a shear transformation in the x -direction with shearing factor a :

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & a & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & -a & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$

- The shear transformation in the y -direction with shearing factor b is

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ b & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -b & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$

- ▶ In space, one can push in two coordinate axis directions and keep the third one fixed.
- ▶ The following is the shear transformation in both x - and y -directions with shearing factors a and b , respectively, keeping the z -coordinate the same

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & a & 0 \\ 0 & 1 & b & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \quad \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -a & 0 \\ 0 & 1 & -b & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix}$$

Projective Transformations

- * Combination of Affine transformations, and Projective warps
- * Properties:
 - * Origin does not necessarily map to origin
 - * Lines map to lines
 - * Parallel lines do not necessarily remain parallel



$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

Projective transformations

- ▶ Projective transformations are so general, little information is necessarily preserved by them.
- ▶ Hyperplanes still map to hyperplanes.
- ▶ Parallelism is not preserved (this is what we would expect, since the perspective transformation can map squares to trapezoids, 'horizontal' lines meeting as they recede with distance).
- ▶ Although ratios are not preserved.
- ▶ In 2D, we associate the point $(\mathbf{a}, \mathbf{b})$ with a set of points in 3D homogeneous coordinates $(w\mathbf{a}, w\mathbf{b}, w)$ for all reals w .

Projective transformations

- ▶ Projective transformations are the most general "linear" transformations and require the use of homogeneous coordinates.
- ▶ Given a point in space in homogeneous coordinate (x,y,z,w) and its image under a projective transform (x',y',z',w') , a projective transform has the following form:

$$\begin{bmatrix} x' \\ y' \\ z' \\ w' \end{bmatrix} = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \\ p_{41} & p_{42} & p_{43} & p_{44} \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \quad \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \\ p_{41} & p_{42} & p_{43} & p_{44} \end{bmatrix}^{-1} \cdot \begin{bmatrix} x' \\ y' \\ z' \\ w' \end{bmatrix}$$

- ▶ 4-by-4 matrices must be non-singular (*i.e.*, invertible). Therefore, projective transformations are more general than affine transformations because the fourth row does not have to contain 0, 0, 0 and 1.
- ▶ Projective transformation can bring finite points to infinity and points at infinity to finite range